BPM501
ER Module
IMCQ 1 - KEY

Spring 2026

1. A 27-year-old man was brought to the physician because of a 1-day history of nausea, and dizziness and a 2-month history of intermittent headaches with changes to his vision. Brain imaging studies show a mass in the sella turcica with inferior extension into the sphenoidal sinus. A biopsy shows that the tumor originated from cells with rounded nuclei. Immunofluorescence shows that the affected cells were positive for GFAP. Which of the following is most likely associated with the cells forming the tumor?

A. Classified as folliculostellate cells
B. Lines colloid filled follicles
C. Forms part of the hypothalamohypohyseal portal system
D. Receives blood supply from the superior hypophyseal artery
E. Releases regulatory hormones into the primary capillary plexus
F. Produces regulatory hormones secreted by PVN and SON
G. Supportive cells of the hypothalamohypophyseal tract

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SOM.MK.I.BPM2.2.ER.1.HCB.1016 Describe the microscopic structure of the posterior pituitary.

A 42-year-old woman comes to the physician because of a 4-month history of secondary amenorrhea and decreased libido and a 6-month history of progressive fatigue, cold intolerance, and un-intentional weight gain of 4.5-kg (10-lb). Her blood pressure is 101/62 mm Hg, pulse is 54/min and respirations are 12/min. Physical examination shows dry skin, bradycardia, and hypotension. Laboratory studies show low levels of TSH, ACTH, LH, and FSH, with corresponding low peripheral hormone levels. An MRI of her brain shows an intact pituitary gland without mass lesions. An arteriogram of the vascular supply of the brain shows sclerotic lesions in the primary capillary plexus of the pituitary gland. A dysfunction of which of the following physiological systems best explains the hormonal abnormalities observed in the laboratory studies?

A. Direct arterial supply to the anterior pituitary
B. Hypothalamo-hypophyseal portal system
C. Posterior pituitary axonal transport system
D. Obstruction of the pituitary venous drainage into the cavernous sinus
E. Systemic circulation-mediated endocrine signaling

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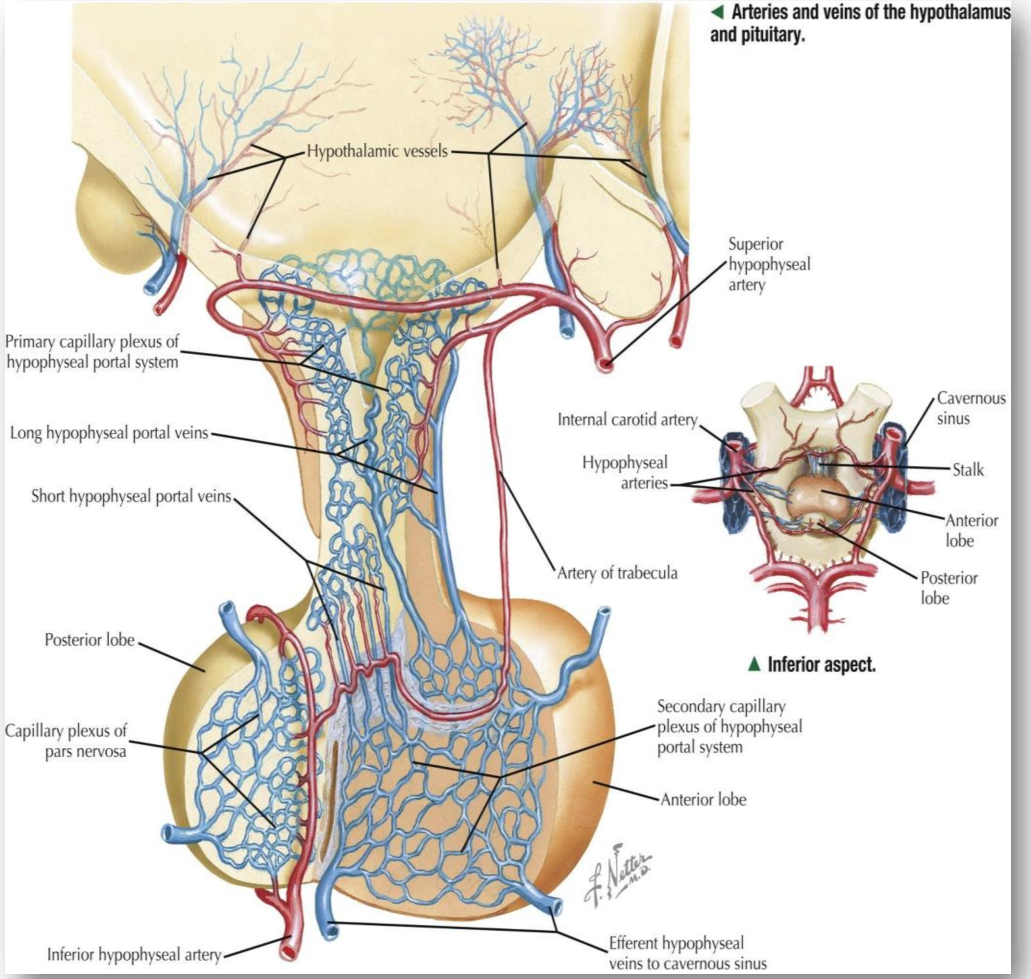
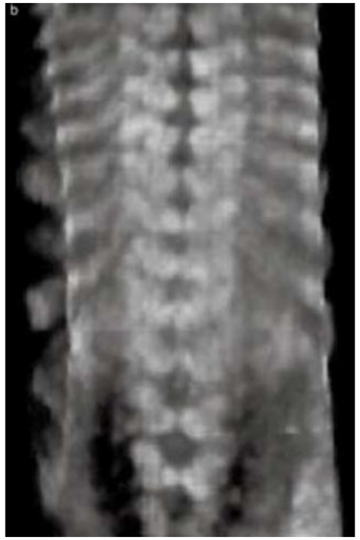
SOM.MK.I.BPM2.2.ER.1.HCB.1009 Discuss the significance of the hypothalamo-hypophyseal portal system in terms of hormone production and secretion.

Hypothalamo-hypophyseal Portal System



3. A 28-year-old G3P1 woman undergoes a routine 18-week fetal ultrasound. The scan shows a fetus with a normal head size but pronounced frontal bossing, rhizomelic shortening of the long bones, and a narrowing of the interpedicular distance in the lumbar spine. An amniocentesis is performed, and genetic analysis of the amniocytes confirms a mutation in the FGFR3 gene. Which of the following terms best describes the type of congenital dysmorphology present in this fetus?

A. Malformation — 20%
B. Sequence — 20%
C. Disruption — 20%
D. Association — 20%
E. Dysplasia — 20%



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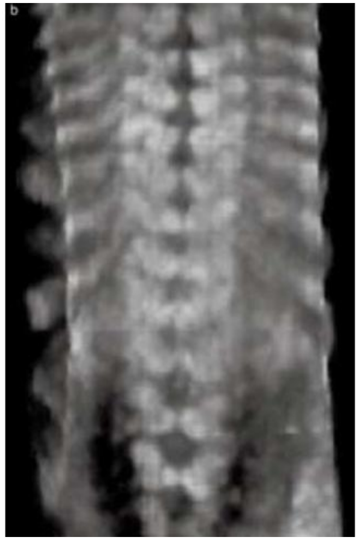

A. Malformation
B. Sequence
C. Disruption
D. Association
✓ E. Dysplasia

SOM.MK.III.BPM2.4.ER.3.GNET.1001 Discuss the vocabulary associated with congenital dysmorphology and give examples.

4. A 60-year-old man comes to the physician for a follow-up examination. He has a 5-year history of type 2 diabetes mellitus and hypertension. He had an episode of myocardial ischemia 1-year ago. His waist to hip ratio is 1.1; BMI is 32 kg/m^2. Which of the following is most likely contributing to his cardiovascular condition?

A. Reduced activity of lipoprotein lipase	0%
B. Increased activity of lipoprotein lipase	0%
C. Reduced secretion of leptin	0%
D. Increased secretion of adiponectin	0%
E. Increased loss of glucose in the urine	0%
F. Low LDL-cholesterol levels	0%

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E.	Increased loss of glucose in the urine	0%
F.	Low LDL-cholesterol levels	0%

SOM.MK.II.BPM2.1.ER.2.BCHM.1037; Evaluate changes in serum lipids (dyslipidemia) in diabetes mellitus and correlate to occurrence of macrovascular changes.

5. A 16-year-old girl is brought to the physician by her parents because of concerns about her lack of menstruation. Her blood pressure is 150/95 mm Hg. Physical examination shows minimal axillary and pubic hair. She has Tanner stage I (prepubertal) breast development. Genital examination shows normal external female genitalia. Results of the laboratory studies are shown. Which of the following is the most likely diagnosis?

Lab test	Patient	Reference range
Hemoglobin	15.2 g/dL	12.0 - 16.0 g/dL
White Blood Cell Count	7,200 cells/µL	4,500 – 11,000 cells/µL
Platelets	250,000/µL	150,000 – 450,000/µL
Blood Urea Nitrogen (BUN)	12 mg/dL	7 – 18 mg/dL
Creatinine	0.8 mg/dL	0.6 – 1.2 mg/dL
Serum sodium (Na^+)	150 mEq/L	135 – 145 mEq/L
Serum potassium (K^+)	3.0 mEq/L	3.5 – 5.0 mEq/L
Fasting glucose	65 mg/dL	70 – 100 mg/dL

A. 3-beta hydroxysteroid dehydrogenase deficiency 0%
B. 11-beta hydroxylase deficiency 0%
C. 17-alpha hydroxylase deficiency 0%
D. 21-alpha hydroxylase deficiency 0%
E. Addison's disease 0%
F. Cushing syndrome due to ACTH oversecretion 0%

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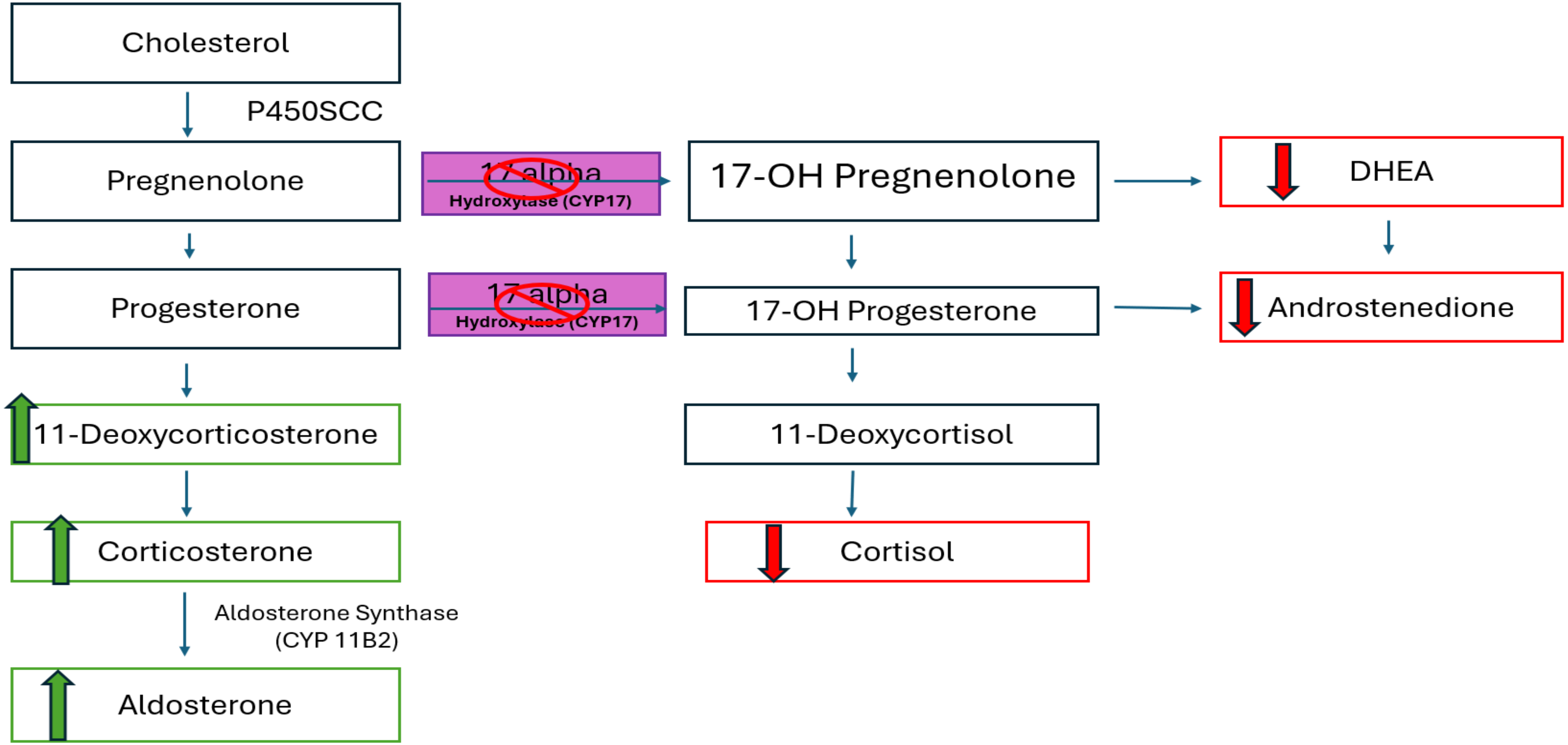
SOM.MK.II.BPM2.1.ER.2.BCHM.1030; Differentiate congenital adrenal hyperplasias (CAH) in patients with deficiencies of 3-hydroxysteroid DH, CYP 21, CYP 11B and CYP 17 based on clinical features and laboratory findings.

CAH: Deficiency of CYP 17 (17a-Hydroxylase)



6. A 55-year-old man comes to the physician because of a 4-month history of unexplained episodic headaches and palpitations. He is admitted to the hospital for evaluation. His blood pressure is 180/125 mmHg, and pulse is 110/min. Laboratory studies from a 24-hour urine test show elevated vanillylmandelic acid (VMA) levels. Elevated levels of which of the following hormones best explain the symptoms in this patient?

A. Dopamine 0%
B. Serotonin 0%
C. Cortisol 0%
D. Epinephrine 0%
E. Melatonin 0%
F. Dehydroepiandrostenedione (DHEA) 0%

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SOM.MK.II.BPM2.1.ER.2.BCHM.1023; Correlate the following disorders to the biochemical basis of the manifestations and the associated laboratory findings: Parkinson disease; Pheochromocytoma; Carcinoid syndrome.

7. A 55-year-old woman comes to the physician because of a 1-month history of difficulty in concentrating on her job, tiredness and increased appetite. She has a 7-year history of hypertension and is on propranolol. Her temperature is 96°F, respirations are 12/min, pulse is 65/min, and blood pressure is 120/70 mmHg. Physical examination shows no abnormalities. Laboratory studies are shown in the table. A deficiency of which of the following is most likely associated with this patient's condition?

Laboratory test	**Patient findings**	**Reference range**
Hemoglobin	13 g/dL	12.0 - 16.0 g/dL
Serum sodium (Na^+)	140 mEq/L	135 – 145 mEq/L
Serum potassium (K^+)	4.5 mEq/L	3.5 – 5.0 mEq/L
TSH	0.8 μU/mL	0.4 -4 μU/mL
T4	10 μg/dL	5-12 μg/dL
T3	45 ng/dL	60-180 ng/dL

A. Thyrotropin releasing hormone 0%
B. T4 5'deiodinase 0%
C. T4 5-deiodinase 0%
D. Thyroglobulin 0%
E. Thyroperoxidase 0%

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✓ B. T4 5'deiodinase 0%

C. T4 5-deiodinase 0%

D. Thyroglobulin 0%

E. Thyroperoxidase 0%

SOM.MK.II.BPM2.1.ER.2.BCHM.1018: Outline and explain the biosynthesis of T3 and T4.

8. A 78-year-old woman comes to the physician because of a 6-month history of progressive difficulty in performing daily activities, such as climbing stairs and rising from a chair. She has no significant past medical history, and her current medications include a multivitamin. Her BMI is 24 kg/m². Physical examination shows decreased muscle mass in her limbs, and her grip strength is notably reduced. Dual-energy X-ray absorptiometry (DEXA) confirms that her appendicular muscle mass is two standard deviations below the mean for young healthy adults. Laboratory findings show normal thyroid function, normal glucose levels, and a slightly reduced serum albumin level. Which of the following interventions is most likely to slow the progression of her condition?

A. Vitamin D supplementation alone 0%

B. Increasing dietary intake of calcium 0%

C. Diet rich in fiber and low in protein 0%

D. High-protein diet and weight training 0%

E. Strict bed rest and limited physical activity 0%

F. Vitamin C supplementation alone 0%

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✔ D. High-protein diet and weight training
E. Strict bed rest and limited physical activity
F. Vitamin C supplementation alone

SOM.MK.III.BPM2.2.ER.3.BCHM.1007; Define frailty and sarcopenia. Review metabolic changes associated with aging highlighting negative nitrogen balance and loss of muscle protein

9. A 9-month-old girl is brought to the emergency department because of a 1-day history of drowsiness, a weak cry, and slow movements. Her blood pressure is 130/90 mmHg, pulse is 105/min, and respirations are 65/min. Physical examination findings and results of laboratory studies are shown. Chromosomal analysis shows 46,XX. A provisional diagnosis of an inherited enzyme deficiency is made. An elevation of which of the following most likely contributes to the patient's blood pressure findings?

Parameter (Serum)	**Patient**	**Reference**
Glucose	40 mg/dL	70-100 mg/dL
Sodium (Na^+)	148 mmol/L	133-146 mmol/L
Potassium (K^+)	2.7 mmol/L	3.5-5.3 mmol/L

A. Aldosterone 0%

B. 11-Deoxycorticosterone 0%

C. 11-Deoxycortisol 0%

D. 17-hydroxy progesterone 0%

E. Dehydroepiandrosterone (DHEA) 0%

F. Androstenedione 0%

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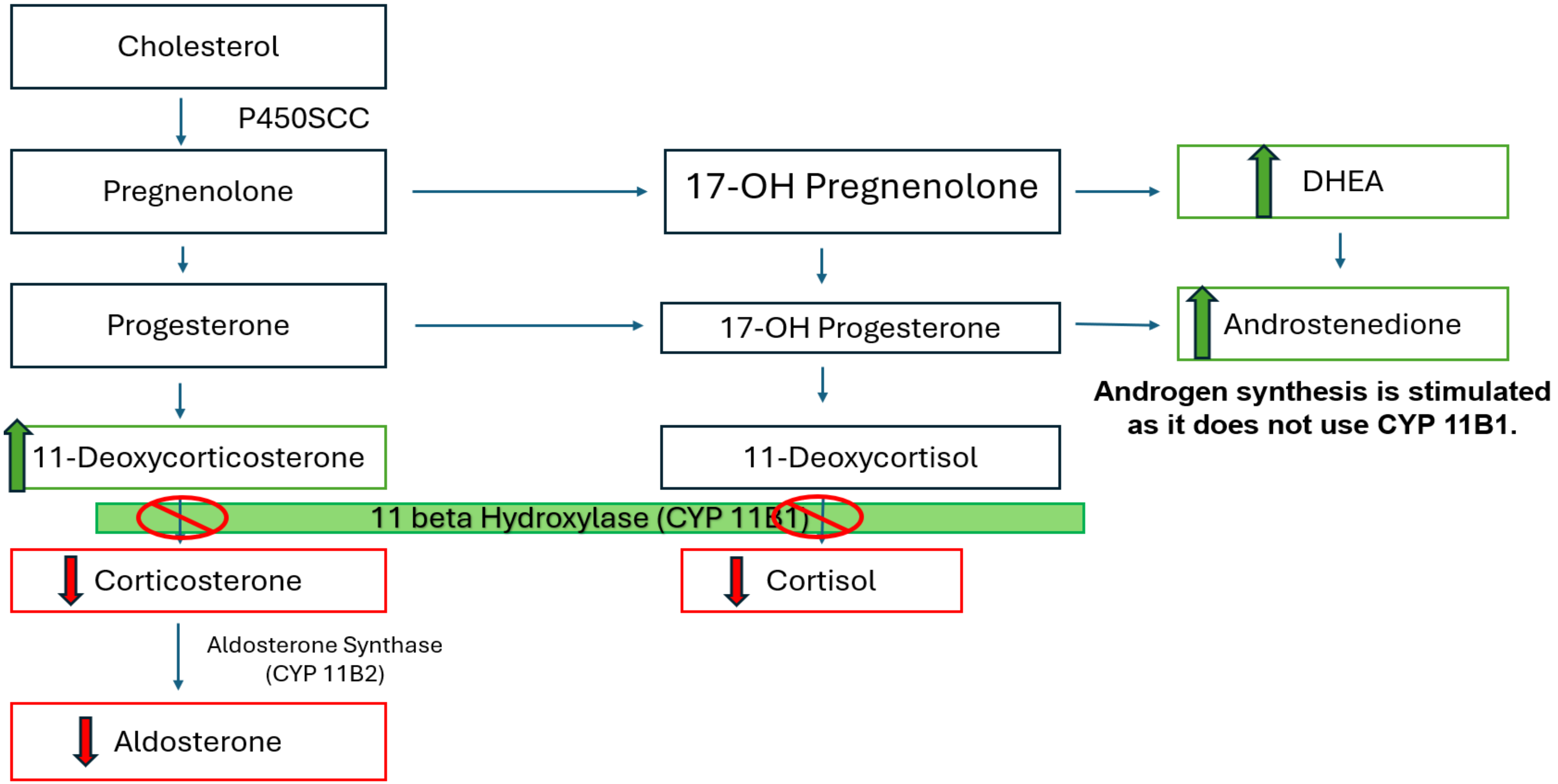
C. 11-Deoxycortisol
D. 17-hydroxy progesterone
E. Dehydroepiandrosterone (DHEA)
F. Androstenedione

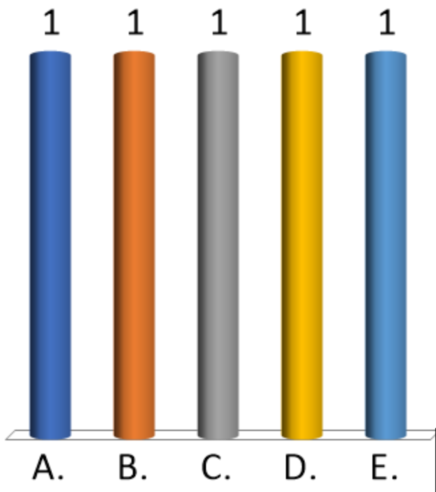
SOM.MK.II.BPM2.1.ER.2.BCHM.1030; Differentiate congenital adrenal hyperplasias (CAH) in patients with deficiencies of 3-hydroxysteroid DH, CYP 21, CYP 11B and CYP 17 based on clinical features and laboratory findings.

CAH: Deficiency of CYP 11B1 (11β-Hydroxylase)



10. A 26-year-old man comes to the clinic because of a 9-month history of a progressive weight gain, difficulty biting food, and prominent cheekbones. He reports a 7-kg (15.4-lb) weight gain. His brother was diagnosed with parathyroid tumor, one year ago. His blood pressure is 156/86 mmHg; other vital signs are within normal limits. Physical examination shows prominent frontal and maxillary bones, protruding mandible, excessive sweating, and coarse skin. Serum studies show increased concentrations of growth hormone and IGF-1. MRI of the brain shows a mass in the sella turcica. Which of the following sets of alterations is most likely in this patient?

Option	GHRH	SRIF	Serum Glucose
A	↓	↓	↓
B	↓	↑	↓
C	↑	↓	↓
D	↑	↑	↑
E	↓	↑	↑



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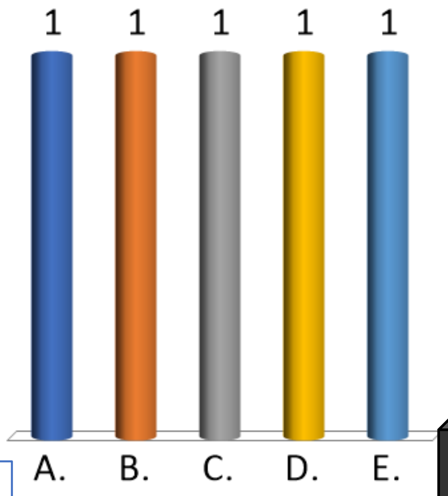
C	↑	↓	↓
D	↑	↑	↑
✔ E	↓	↑	↑



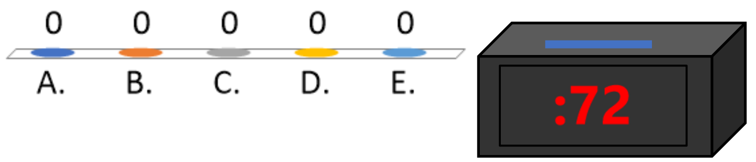
SOM.MK.II.BPM2.1.ER.1.PHYS.1012. Predict the developmental effects of too much or too little growth hormone.

11. A 44-year-old man comes to the physician because of an 8-month history of lethargy, puffiness of face and extremities, constipation, and a 6-kg (13.2-lb) weight gain. His temperature is 36.4°C (97.5°F), pulse is 54/min, and blood pressure is 128/84 mm Hg. Physical examination shows dry skin, non-pitting edema of extremities, sluggish deep tendon reflexes, and loss of scalp hair. Results of laboratory studies are shown:

Test	Patient value (Reference range)
Serum TSH	0.02 mU/L (normal: 0.5-5.0)
Serum free T4	0.4 µg/dL (normal: 4-11.5)
Autoantibodies	Negative for anti-Thyroid Peroxidase (anti-TPO)

Which of the following mechanisms is the most likely cause of this patient’s condition?

A. Primary hypothyroidism
B. Secondary hypothyroidism
C. Iodine deficiency
D. Hashimoto’s thyroiditis
E. Graves disease



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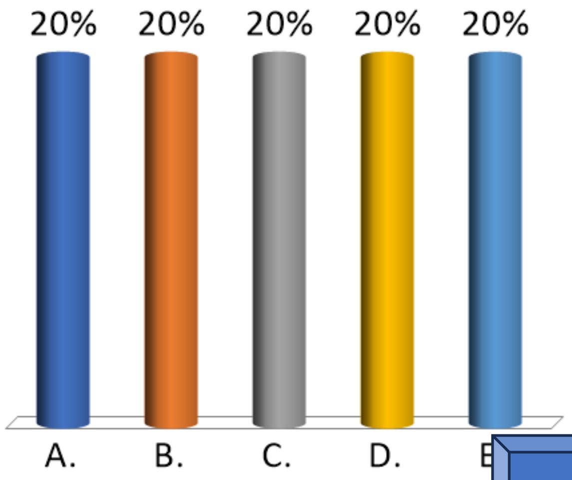
SOM.MK.II.BPM2.1.ER.1.PHYS.1055. List and explain the causes of thyroid gland enlargement. Predict the consequences of thyroid hormone under- and over-secretion.

12. A 49-year-old man comes to the physician because of a 6-month history of nausea, loss of appetite, dizziness, fatigue, and a 4-kg (8.8-lb) weight loss. One year ago, he was diagnosed with autoimmune thyroiditis and is on thyroid supplements. His pulse is 90/min and blood pressure is 86/58 mmHg; other vital signs are within normal limits. Physical examination shows pallor, dry mucous membranes, and hyperpigmentation of tongue, gums, and skin creases. Laboratory studies show low serum concentrations of sodium, glucose, and bicarbonate. Serum hormone studies show decreased concentrations of cortisol and aldosterone but increased ACTH. Which of the following sets of additional serum alterations is most likely in this patient?

Option	Osmolality	K^+	pH
A	↓	↓	Low
B	↓	↑	Low
C	↑	↓	Low
D	↑	↑	High
E	↓	↑	High



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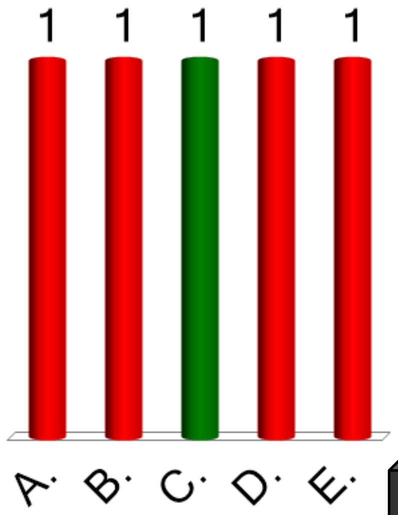
C	↑	↓	Low
D	↑	↑	High
E	↓	↑	High

SOM.MK.II.BPM2.1.ER.1.PHYS.1026. Explain biological actions of cortisol and adrenal androgens. Identify the causes and consequences of over secretion and under secretion of cortisol. SOM.MK.II.BPM2.1.ER.1.PHYS.1029. Identify the causes and consequences of - over secretion and - under secretion of mineralocorticoids -deficiency of enzymes in the steroid synthesis pathway.

13. A 43-year-old man comes to the physician because of a 6-week history of constipation, increased urination, increased thirst, and abdominal pain. Three months ago, he underwent surgery to remove a brain tumor. His blood pressure is 110/68 mmHg; other vital signs are within normal limits. Physical examination shows dry tongue, increased skin turgor, and sunken eyes. Serum studies show elevated sodium and osmolarity. Serum glucose, potassium and bicarbonate concentrations are within reference ranges. Urinalysis shows increased urine volume but reduced osmolarity. Ultrasonography of the abdomen shows no abnormalities. Water deprivation for 6 hours does not alter urine osmolarity, but administration of synthetic vasopressin increases urine osmolarity. Which of the following mechanisms is the most likely cause of this patient's condition?

A. Defective functioning of V2 receptors in kidneys
B. Excess release of vasopressin from the posterior pituitary
C. Reduced hypothalamic vasopressin secretion
D. Reduced pancreatic insulin secretion
E. Increased activity of aquaporin2 channels in the renal tubules



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SOM.MK.II.BPM2.1.ER.1.PHYS.1021. Identify disease states caused by a) over secretion, and b) under secretion of vasopressin and list the principal symptoms of each.

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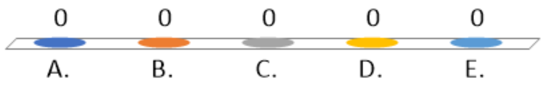
14. A 37-year-old woman comes to the physician because of a 6-week history of recurrent headaches and muscle cramps. Her blood pressure is 178/98 mm Hg; rest of the vital signs are within normal limits. Physical examination shows loud second heart sound in the right upper sternal border. Results of laboratory studies are shown:

Test	Patient's test results compared to reference values
Serum hormones	**High aldosterone, Low renin**
Serum electrolytes	**High sodium, Low potassium**
Acid-base status	**High serum pH, High serum bicarbonate**

A CT scan of the abdomen shows a mass in the upper pole of the left kidney. Which of the following is the most likely diagnosis in this patient?

A. Polycystic kidney disease on left side
B. Left renal artery stenosis
C. Secondary hyperaldosteronism
D. Primary hyperaldosteronism
E. ACTH-independent Cushing syndrome



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14. A 37-year-old woman comes to the physician because of a 6-week history of recurrent headaches and muscle cramps. Her blood pressure is 178/98 mm Hg; rest of the vital signs are within normal limits. Physical examination shows loud second heart sound in the right upper sternal border. Results of laboratory studies are shown:

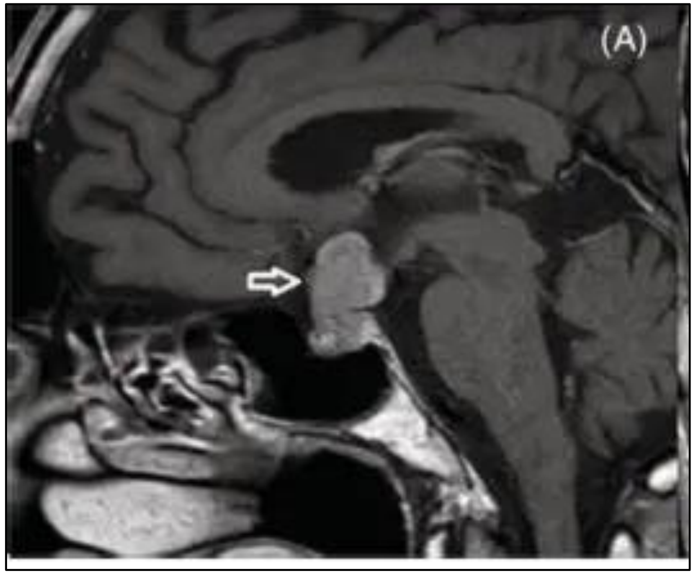
Test	Patient's test results compared to reference values
Serum hormones	**High aldosterone, Low renin**
Serum electrolytes	**High sodium, Low potassium**
Acid-base status	**High serum pH, High serum bicarbonate**

A CT scan of the abdomen shows a mass in the upper pole of the left kidney. Which of the following is the most likely diagnosis in this patient?

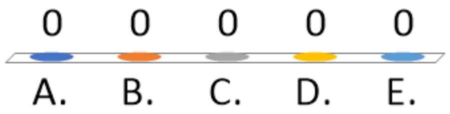
A. Polycystic kidney disease on left side
B. Left renal artery stenosis
C. Secondary hyperaldosteronism
✓D. Primary hyperaldosteronism
E. ACTH-independent Cushing syndrome

SOM.MK.II.BPM2.1.ER.1.PHYS.1029. Identify the causes and consequences of over secretion and under secretion of mineralocorticoids

15. A 41-year-old man comes to the physician because of a 10-month history of inability to sustain an erection of sufficient rigidity and duration for sexual intercourse, loss of libido, recurrent headaches, and blurred vision. Vital signs are within normal limits. Physical examination shows loss of temporal field of vision in both eyes. Serum studies show reduced concentrations of testosterone and gonadotropins (both LH and FSH). Semen study shows low sperm count. MRI of the brain is shown; the arrow indicates a mass in the pituitary fossa. Elevated levels of which of the following hormones best explains this patient's infertility?



A. Dopamine
B. Cortisol
C. ACTH
D. Prolactin
E. GnRH



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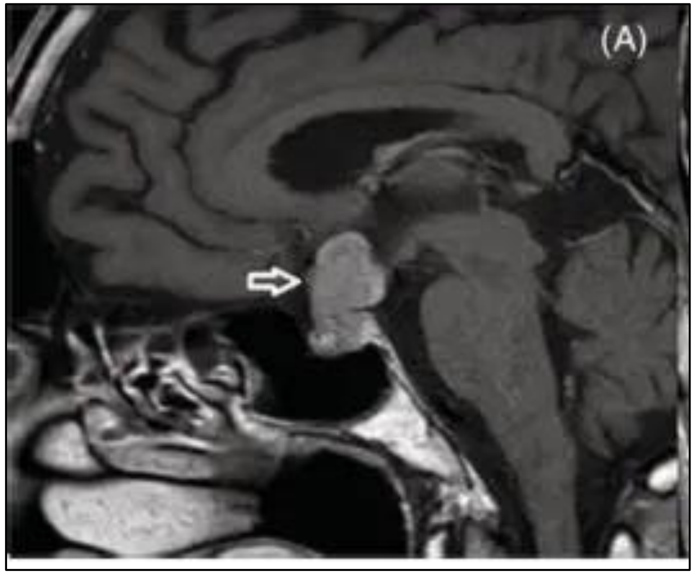
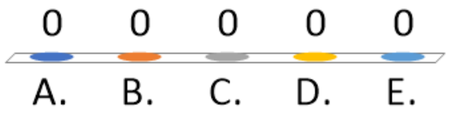


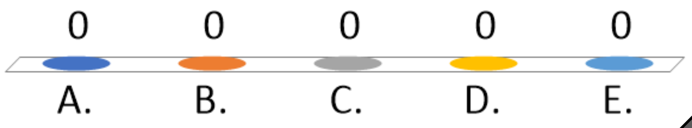
A. Dopamine
B. Cortisol
C. ACTH
D. Prolactin
E. GnRH

SOM.MK.II.BPM2.1.ER.1.PHYS.1017. Describe symptoms of prolactinomas and prolactin imbalance.



16. A 34-year-old woman comes to the physician because of a 4-month history of recurrent episodes of palpitations, nervousness, heat intolerance and a 4-kg (8.8-lb) weight loss. Her temperature is 37.8°C (100°F), pulse is 124/min and, blood pressure is 146/62 mm Hg. Physical examination shows diaphoresis, exophthalmos, and fine tremors of the stretched hands. The neck examination shows a non-tender, enlarged thyroid gland. Laboratory studies show serum TSH of 0.04 mU/L (normal: 0.3-5.0), serum thyroxine of 17.6 µg/dL (normal:4-11.5), and positive autoantibodies. Which of the following best explains an increased pulse rate in this patient?

A. Increased beta1 receptor sensitivity
B. Increased blood pressure
C. Increased blood flow to skin
D. Secretion of catecholamines from adrenal medulla
E. Increased Na+/K+-ATPase activity



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C. Increased blood flow to skin
D. Secretion of catecholamines from adrenal medulla
E. Increased Na+/K+-ATPase activity

SOM.MK.II.BPM2.1.ER.1.PHYS.1053. Describe the physiologic effects and mechanisms of action of thyroid hormones and clinical applications.